



## Thermodynamics ENME601

### Mid-Semester Test

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## Semester 2, 2025

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**TIME ALLOWED:** 2 hour and 30 minutes

**MARKS:** TOTAL: 30 marks

**INSTRUCTIONS:** Answer **ALL THREE** questions.

All answers except diagrams and sketches **must be in ink**.

Write your answers in the **answer booklet**.

Draft sheets will be supplied but will **NOT** be evaluated.

**Exam conditions apply!**

**NOTES:**

A paper-based, open-book exam where **ALL** handwritten or printed notes are allowed.

Access to the internet and use of electronic devices are **NOT** allowed, except for the use of a calculator.

Candidates should show **all working** in calculations.

**Q1)** A 10 kg system was taken from 25°C to 75°C. Given the specific heat capacity of the system is  $c_p = 3000 \text{ J/(kg} \cdot ^\circ\text{C)}$ . (9 marks in total)

(a) How much energy in the form of heat was added to the system to produce this temperature increase? (4 marks)

(b) If the system received the heat over a period of 20 seconds, calculate the heat transfer rate  $\dot{Q}$  in kW. (3 marks)

(c) For the same system, if sufficient heat were supplied, its temperature would rise from 25 °C to 80 °C without heat loss. In reality, because of heat losses, the system only reached 75 °C. How much heat was lost during the process? (2 marks)

**Q2)** 2 kg of water that is initially at 95°C with a quality of 0.35 occupies a linear spring-loaded piston–cylinder device. This device is now heated until the pressure rises to 800 kPa and the temperature is 275°C. To determine: (12 marks in total)

(a) The specific volume ( $v_o$ ) at the initial state. (3 marks)

(b) The specific volume ( $v_1$ ) at the final state. (4 marks)

(c) The total work produced during this process, in kJ. (5 marks)

**Q3)** Refrigerant-134a at 0.8 MPa and 90°C is to be cooled to 0.8 MPa and 30°C in a condenser by the air (*can be treated as ideal gas*). The air enters at 100 kPa and 22°C with a volume flow rate of 400 m<sup>3</sup>/min and leaves at 95 kPa and 57°C. To determine: (9 marks in total)

(a) The enthalpy ( $h_1$ ) of R134a at the inlet. (2 marks)

(b) The enthalpy ( $h_2$ ) of R134a at the outlet. (2 marks)

(c) The enthalpy ( $h_3$ ) of air at the inlet. (1 mark)

(d) The enthalpy ( $h_4$ ) of air at the outlet. (1 mark)

(e) The mass flow rate of the refrigerant. (3 marks)